

Replicating ‘Estimates of the Trade and Welfare Effects of NAFTA Get access Arrow’ by Caliendo and Parro (2015)

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1 Introduction

Caliendo and Parro (2015) evaluate the economic impacts of the North American Free Trade Agreement using a multi-sector Ricardian trade model incorporating intermediate goods and input-output linkages. Their findings show that Mexico’s welfare increases by 1.31%, USA’s welfare increases by 0.08%, and Canada’s welfare declines by 0.06% while at the same time intra-bloc trade increases by 118% for Mexico, 41% for the USA and 11% for Canada.

This replication effort uses Vargas Sepulveda (2025) to leverage Armadillo’s cube class (quasi third-order tensor, or 3D-matrix), which simplifies solving non-linear systems as the ones required for the Ricardian model with trade sectors.

The tables below correspond to an independent simulation of the results from section 5.1. in Caliendo and Parro (2015).

```
library(cp2015)
library(data.table)
options(knitr.kable.NA = "")
```

The `cp2015` package (<https://github.com/pachadotdev/cp2015/blob/main/README.md>) includes `cp2015nafta`, a list containing the required model sets, intermediate consumption, final consumption, value added, bilateral trade and tariff data, deficits, and trade elasticities.

For different datasets, such as those with trade figures from the OECD, it is required to reshape the dataset before passing it to the package functions as in the structure below:

```
str(cp2015nafta)
```

List of 7

```
$ sets                                :List of 2
..$ regions: chr [1:31] "Argentina" "Australia" "Austria" "Brazil" ...
..$ sectors: chr [1:40] "Agriculture" "Mining" "Food" "Textile" ...
$ intermediate_consumption:Classes 'data.table' and 'data.frame': 49600 obs. of 4 variables
..$ input : chr [1:49600] "Agriculture" "Agriculture" "Agriculture" "Agriculture" ...
..$ sector: chr [1:49600] "Agriculture" "Agriculture" "Agriculture" "Agriculture" ...
..$ region: chr [1:49600] "Argentina" "Australia" "Austria" "Brazil" ...
..$ value : num [1:49600] 2.62e+09 2.96e+09 5.20e+08 3.89e+09 7.63e+09 ...
..- attr(*, ".internal.selfref")=<pointer: (nil)>
$ final_consumption                  :Classes 'data.table' and 'data.frame': 1240 obs. of 3 variables
..$ sector: chr [1:1240] "Agriculture" "Agriculture" "Agriculture" "Agriculture" ...
..$ region: chr [1:1240] "Argentina" "Australia" "Austria" "Brazil" ...
..$ value : num [1:1240] 7.16e+09 6.48e+09 4.76e+09 1.59e+10 4.74e+09 ...
..- attr(*, ".internal.selfref")=<pointer: (nil)>
$ value_added                        :Classes 'data.table' and 'data.frame': 1240 obs. of 3 variables
..$ sector: chr [1:1240] "Agriculture" "Agriculture" "Agriculture" "Agriculture" ...
..$ region: chr [1:1240] "Argentina" "Australia" "Austria" "Brazil" ...
..$ value : num [1:1240] 1.24e+10 1.18e+10 5.12e+09 2.83e+10 1.51e+10 ...
..- attr(*, ".internal.selfref")=<pointer: (nil)>
$ trade                              :Classes 'data.table' and 'data.frame': 38440 obs. of 9 variables
..$ sector      : chr [1:38440] "Agriculture" "Agriculture" "Agriculture" "Agriculture" ...
..$ exporter    : chr [1:38440] "Argentina" "Australia" "Austria" "Brazil" ...
..$ importer    : chr [1:38440] "Argentina" "Argentina" "Argentina" "Argentina" ...
..$ value       : num [1:38440] 1.91e+10 2.19e+06 5.56e+04 8.87e+07 1.50e+06 ...
..$ tariff      : num [1:38440] 0 0.0417 0.0408 0.0488 0.0288 ...
```

```

..$ tariff_bln: num [1:38440] 0 0.0417 0.0408 0.0488 0.0288 ...
..$ tariff_cfl: num [1:38440] 0 0.0417 0.0408 0.0488 0.0288 ...
..$ d_bln      : num [1:38440] 1 1 1 1 1 1 1 1 1 1 ...
..$ d_cfl      : num [1:38440] 1 1 1 1 1 1 1 1 1 1 ...
..- attr(*, ".internal.selfref")=<pointer: (nil)>
$ deficit      :Classes 'data.table' and 'data.frame': 31 obs. of 2 variables:
..$ region: chr [1:31] "Argentina" "Australia" "Austria" "Brazil" ...
..$ D      : num [1:31] 3.51e+09 -4.19e+09 1.62e+10 -1.43e+10 -1.07e+10 ...
..- attr(*, ".internal.selfref")=<pointer: (nil)>
$ theta        :Classes 'data.table' and 'data.frame': 40 obs. of 2 variables:
..$ sector: chr [1:40] "Agriculture" "Mining" "Food" "Textile" ...
..$ value : num [1:40] 9.11 13.53 2.62 8.1 11.5 ...
..- attr(*, ".internal.selfref")=<pointer: (nil)>

```

Caliendo and Parro (2015) impose zero aggregate deficits in the baseline and counterfactual scenarios. We use `zero_aggregate_deficit = TRUE` to reproduce that scenario.

```
results <- run_cp2015(data = cp2015nafta, zero_aggregate_deficit = TRUE)
```

Solving the baseline.

```

Iteration: 1 - ||F(x)||: 4075219334
Iteration: 100 - ||F(x)||: 0.009735903192
Iteration: 200 - ||F(x)||: 0.001738052373
Iteration: 300 - ||F(x)||: 0.0003572560362
Iteration: 400 - ||F(x)||: 6.213691551e-05
Iteration: 500 - ||F(x)||: 1.315719258e-05
Iteration: 600 - ||F(x)||: 2.223333097e-06
Iteration: 700 - ||F(x)||: 4.8383701e-07
Iteration: 796 - ||F(x)||: 9.686823042e-08

```

Solving the counterfactual.

```

Iteration: 1 - ||F(x)||: 4074074133
Iteration: 100 - ||F(x)||: 0.01007323017
Iteration: 200 - ||F(x)||: 0.001681247597
Iteration: 300 - ||F(x)||: 0.0003469702056
Iteration: 400 - ||F(x)||: 5.838458239e-05
Iteration: 500 - ||F(x)||: 1.212108083e-05
Iteration: 600 - ||F(x)||: 2.028661454e-06
Iteration: 700 - ||F(x)||: 4.233016861e-07
Iteration: 779 - ||F(x)||: 9.651468184e-08

```

The welfare decomposition includes terms of trade, volume of trade, and technical efficiency. The latter is zero in this tariff-only example. From the model results, we need to aggregate its outputs in different ways to replicate tables two to five.

2 Welfare effects from NAFTA's tariff reductions (Table 2)

```
nafta <- c("Mexico", "Canada", "USA")
welfare <- results$welfare[region %in% nafta]
welfare[, tech := NULL]
welfare <- welfare[match(nafta, region), ]

knitr::kable(
  welfare,
  digits = 2,
  latex = TRUE,
  col.names = c("Country", "Total", "Terms of trade", "Volume of trade", "Real wages"),
  caption = "Welfare effects from NAFTA's tariff reductions (Table 2)"
)
```

Table 1: Welfare effects from NAFTA's tariff reductions (Table 2)

Country	Total	Terms of trade	Volume of trade	Real wages
Mexico	1.31	-0.41	1.72	1.72
Canada	-0.06	-0.11	0.04	0.32
USA	0.08	0.04	0.04	0.11

3 Bilateral welfare effects from NAFTA's tariff reductions (Table 3)

```
tot_bilateral <- results$tot[region %in% nafta]
tot_bilateral[, partner := fifelse(partner %in% nafta, "NAFTA", "ROW")]
tot_bilateral <- tot_bilateral[, .(ToT = sum(tot)), by = .(region, partner)]

vot_bilateral <- results$vot[region %in% nafta]
vot_bilateral[, partner := fifelse(partner %in% nafta, "NAFTA", "ROW")]
vot_bilateral <- vot_bilateral[, .(VoT = sum(vot)), by = .(region, partner)]

bilateral <- merge(
  tot_bilateral,
  vot_bilateral,
  by = c("region", "partner"),
  all = TRUE,
  sort = FALSE
)
```

```

)

bilateral <- dcast(
  bilateral,
  region ~ partner,
  value.var = c("ToT", "VoT")
)

bilateral <- bilateral[match(nafta, region), ]

knitr::kable(
  bilateral,
  digits = 2,
  latex = TRUE,
  col.names = c("Country", "ToT NAFTA", "ToT ROW", "VoT NAFTA", "VoT ROW"),
  caption = "Bilateral welfare effects from NAFTA's tariff reductions
    (Table 3)"
)

```

Table 2: Bilateral welfare effects from NAFTA's tariff reductions (Table 3)

Country	ToT NAFTA	ToT ROW	VoT NAFTA	VoT ROW
Mexico	-0.39	-0.02	1.80	-0.08
Canada	-0.09	-0.02	0.08	-0.04
USA	0.03	0.01	0.04	0.00

4 Sectoral contribution to welfare effects from NAFTA's tariff reductions (Table 4)

```

tot_sector <- results$tot[region %in% nafta]
tot_sector <- tot_sector[, .(ToT = sum(tot)), by = .(region, sector)]
tot_sector[, ToT := 100 * ToT / sum(ToT), by = region]

vot_sector <- results$vot[region %in% nafta]
vot_sector <- vot_sector[, .(VoT = sum(vot)), by = .(region, sector)]
vot_sector[, VoT := 100 * VoT / sum(VoT), by = region]

sectoral <- merge(

```

```

    tot_sector,
    vot_sector,
    by = c("region", "sector"),
    all = TRUE,
    sort = FALSE
  )

sectoral <- dcast(
  sectoral,
  sector ~ region,
  value.var = c("ToT", "VoT")
)

setcolorder(
  sectoral,
  c("sector", "ToT_Mexico", "VoT_Mexico", "ToT_Canada", "VoT_Canada",
    "ToT_USA", "VoT_USA")
)

knitr::kable(
  sectoral,
  digits = 2,
  latex = TRUE,
  col.names = c("Sector", gsub("_", " ", colnames(sectoral)[-1])),
  caption = "Sectoral contribution to welfare effects from NAFTA's
    tariff reductions (Table 4)"
)

```

Table 3: Sectoral contribution to welfare effects from NAFTA's tariff reductions (Table 4)

Sector	ToT Mexico	VoT Mexico	ToT Canada	VoT Canada	ToT USA	VoT USA
Agriculture	-0.13	2.87	3.41	-0.01	3.41	0.65
Air Transport	0.00	0.00	0.00	0.00	0.00	0.00
Auto	13.79	4.78	29.50	27.77	15.85	4.47
Aux Transport	0.00	0.00	0.00	0.00	0.00	0.00
Basic metals	1.07	3.02	10.08	1.48	3.40	1.05
Chemicals	0.57	2.15	5.74	0.08	5.60	1.11
Communication	20.97	3.64	2.67	0.15	11.61	4.58
Computer	0.00	0.00	0.00	0.00	0.00	0.00
Construction	0.00	0.00	0.00	0.00	0.00	0.00

Sector	ToT Mexico	VoT Mexico	ToT Canada	VoT Canada	ToT USA	VoT USA
Education	0.00	0.00	0.00	0.00	0.00	0.00
Electrical	41.20	25.77	1.37	7.18	24.19	42.22
Electricity	0.00	0.00	0.00	0.00	0.00	0.00
Finance	0.00	0.00	0.00	0.00	0.00	0.00
Food	0.45	1.17	3.56	2.37	3.16	1.04
Health	0.00	0.00	0.00	0.00	0.00	0.00
Hotels	0.00	0.00	0.00	0.00	0.00	0.00
Land Transport	0.00	0.00	0.00	0.00	0.00	0.00
Machinery n.e.c	3.68	4.32	5.16	-0.02	5.63	0.65
Medical	4.72	1.34	0.94	-0.23	3.48	4.46
Metal products	0.90	5.56	2.22	7.99	1.61	1.06
Minerals	0.05	0.73	0.93	0.47	0.70	0.57
Mining	-3.01	0.25	4.04	-0.20	1.54	0.04
Office	8.37	4.72	2.32	-0.82	3.50	1.43
Other	2.63	1.92	0.81	-0.10	2.90	1.69
Other Business	0.00	0.00	0.00	0.00	0.00	0.00
Other Transport	0.21	0.82	12.93	-0.97	1.51	0.32
Other services	0.00	0.00	0.00	0.00	0.00	0.00
Paper	0.39	3.82	5.86	0.49	2.83	0.33
Petroleum	-0.09	14.61	0.60	30.41	1.85	11.37
Plastic	0.62	4.21	2.53	7.56	1.61	0.32
Post	0.00	0.00	0.00	0.00	0.00	0.00
Private	0.00	0.00	0.00	0.00	0.00	0.00
Public	0.00	0.00	0.00	0.00	0.00	0.00
R&D	0.00	0.00	0.00	0.00	0.00	0.00
Real State	0.00	0.00	0.00	0.00	0.00	0.00
Renting Mach	0.00	0.00	0.00	0.00	0.00	0.00
Retail	0.00	0.00	0.00	0.00	0.00	0.00
Textile	3.30	12.03	1.15	16.19	4.32	22.24
Water Transport	0.00	0.00	0.00	0.00	0.00	0.00
Wood	0.30	2.26	4.17	0.24	1.31	0.41

5 Trade effects from NAFTA's tariff reductions (Table 5)

```
trade <- results$trade
trade <- trade[importer %in% nafta & exporter %in% nafta]
trade <- trade[, .(trade_bln = sum(trade_bln), trade_cfl = sum(trade_cfl)),
```

```

    by = .(importer, exporter)
  ]
  trade[, change := (trade_cfl / trade_bln - 1) * 100]
  trade[exporter == importer, change := NA_real_]

trade <- dcast(
  trade,
  importer ~ exporter,
  value.var = "change"
)

trade <- trade[match(nafta, importer), ]
setcolororder(trade, c("importer", "Mexico", "Canada", "USA"))

knitr::kable(
  trade,
  digits = 2,
  latex = TRUE,
  col.names = c("Importer/Exporter", nafta),
  caption = "Trade effects from NAFTA's tariff reductions (Table 5)"
)

```

Table 4: Trade effects from NAFTA's tariff reductions (Table 5)

Importer/Exporter	Mexico	Canada	USA
Mexico		116.60	118.31
Canada	58.57		9.49
USA	109.54	6.57	

References

- Caliendo, Lorenzo, and Fernando Parro. 2015. "Estimates of the Trade and Welfare Effects of NAFTA." *The Review of Economic Studies* 82 (1): 1–44.
- Vargas Sepulveda, Mauricio. 2025. "Armadillo4r: An R Package to Use the Armadillo C++ Library." *Available at SSRN 5848422*.